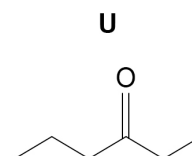
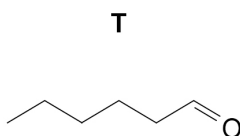
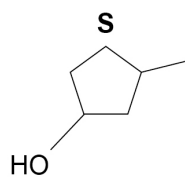
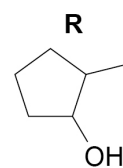
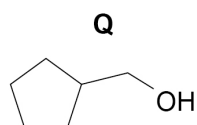
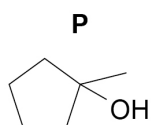


QQ6 Alcohol

AQA 2020,2021,2022

0 3

This question is about the structural isomers shown.



0 3 . 1

Identify the isomer(s) that would react when warmed with acidified potassium dichromate(VI).

State the expected observation when acidified potassium dichromate(VI) reacts.

[2 marks]

Isomer(s) _____

Expected observation _____



0 3 . 3

Separate samples of each isomer are warmed with ethanoic acid and a few drops of concentrated sulfuric acid. In each case the mixture is then poured into a solution of sodium hydrogencarbonate.

Identify the isomer(s) that would react with ethanoic acid.

Suggest a simple way to detect if the ethanoic acid reacts with each isomer.

Give a reason why the mixture is poured into sodium hydrogencarbonate solution.

[3 marks]

Isomer(s) _____

Suggestion _____

Reason _____

0 3 . 4

State the type of structural isomerism shown by isomers **P**, **Q**, **R** and **S**.

[1 mark]

0 3 . 5

Describe fully how infrared spectra can be used to distinguish between isomers **R**, **S** and **T**.

Use data from **Table A** in the Data Booklet in your answer.

[4 marks]

0 3 . 6

State why mass spectrometry using electrospray ionisation is **not** a suitable method to distinguish between the isomers.

[1 mark]



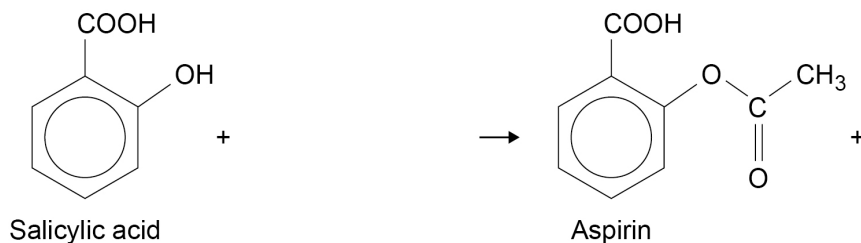
Question	Answers	Additional comments/Guidelines	Mark
03.1	M1 Q, R, S, T	M1 Allow the mark for candidates who correctly name or draw the isomers.	1
	M2 (Orange solution) turns green	Independent	1
03.3	M1 P, Q, R, S	As above	1
	M2 Sweet smelling (liquid)		1
	M3 To react with (remove excess) acid / neutralise	Allow easier to identify the smell	1
03.4	Position	Allow positional	1

03.5	M1 R & S have an <u>O-H alcohols</u> peak at <u>3230-3550</u> cm^{-1} M2 T has <u>C=O</u> peak at <u>1680-1750</u> cm^{-1} M3 R & S (unique) fingerprint region or below 1500 cm^{-1} M4 Compare to a database / known spectra (and look for an exact match)	Allow value within the range	1 1 1 1
03.6	All have the same M_r	Allow same (molecular) ion M/Z peak same molecular formula	1

0 4 . 4

Complete the equation for the reaction of salicylic acid with ethanoic anhydride to produce aspirin.

[1 mark]



0 4 . 5

A 6.01 g sample of salicylic acid ($M_r = 138.0$) is reacted with 10.5 cm^3 of ethanoic anhydride ($M_r = 102.0$). In the reaction the yield of aspirin is 84.1%

The density of ethanoic anhydride is 1.08 g cm^{-3}

Show by calculation which reagent is in excess.

Calculate the mass, in g, of aspirin ($M_r = 180.0$) produced.

[5 marks]

Reagent in excess _____

Mass of aspirin _____ g

Turn over ►



0 4 . 6

Suggest **two** ways in which the melting point of the crude aspirin collected in step 9 would differ from the melting point of pure aspirin.

[2 marks]

Difference 1 _____

Difference 2 _____

0 4 . 7

The crude aspirin can be purified by recrystallisation using hot ethanol (boiling point = 78 °C) as the solvent.

Describe **two** important precautions when heating the mixture of ethanol and crude aspirin.

[2 marks]

Precaution 1 _____

Precaution 2 _____

0 4 . 8

The pure aspirin is filtered under reduced pressure. A small amount of cold ethanol is then poured through the Buchner funnel.

Explain the purpose of adding a small amount of cold ethanol.

[1 mark]

0 4 . 9

A sample of the crude aspirin is kept to compare with the purified aspirin.

Describe **one** difference in appearance you would expect to see between these two solid samples.

[1 mark]



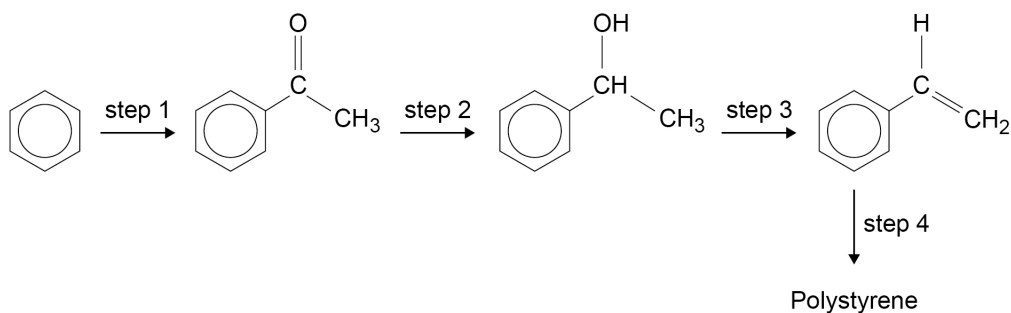
Question	Answers	Additional comments/Guidelines	Mark
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04.4	LHS + (CH ₃ CO) ₂ O RHS + CH ₃ COOH		1
04.5	M1 Amount salicylic acid = $\frac{6.01}{138} = 4.36 \times 10^{-2}$ mol M2 Mass (CH₃CO)₂O = $10.5 \times 1.08 = 11.34$ g M3 Amount (CH₃CO)₂O = $\frac{11.34}{102} = 1.11 \times 10^{-1}$ mol M4 (CH₃CO)₂O is in excess M5 Mass aspirin = $M1 \times 0.841 \times 180 = 6.59$ g	Allow conseq from wrong mole ratio in 04.4 Must show and state that ethanoic anhydride is in excess For M4/M5 ecf from M1/M3 Allow 2 sf or more.	1 1 1 1 1

04.6	M1 Value lower		1
	M2 Range of values	For M2 allow mpt not sharp or a larger range of melting points	1
04.7	M1 (Ethanol is flammable so) use a water bath to heat / do not use a Bunsen burner	Must give practical step, not just state hazard	1
	M2 Heat to temp below bp (so ethanol does not boil away)	Allow use min vol solvent	1
04.8	To remove any soluble impurities	Allow To avoid aspirin dissolving (small amount cold solvent used) Allow To remove/(wash away) any ethanolic solution on the product.	1
04.9	Pure product will have (larger) crystals / needle-like crystals / lighter in colour	Allow whiter, less grey, more crystalline, less powdery, shinier, single colour Must be tied to pure product Allow opposite points tied to the crude product	1

0 6

Polystyrene can be made from benzene in the series of steps shown.

**0 6 . 1**

State the type of reaction in step 1.

Identify the reagent(s) and conditions needed for step 1.

[3 marks]

Type of reaction _____

Reagent(s) _____

Conditions _____

0 6 . 2

State the name of the mechanism for the reaction in step 2.

Identify the inorganic reagent needed for step 2.

Name the organic product of step 2.

[3 marks]

Name of mechanism _____

Inorganic reagent _____

Name of organic product _____



0 6 . 3 The organic product of step **2** is reacted with concentrated sulfuric acid in step **3**.

Outline the mechanism for step **3**.

[3 marks]

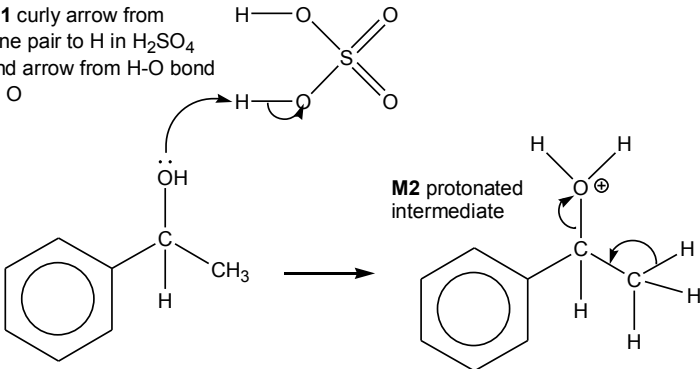
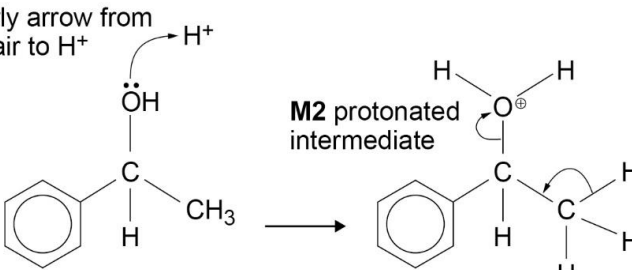
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Turn over for the next question

Turn over ►

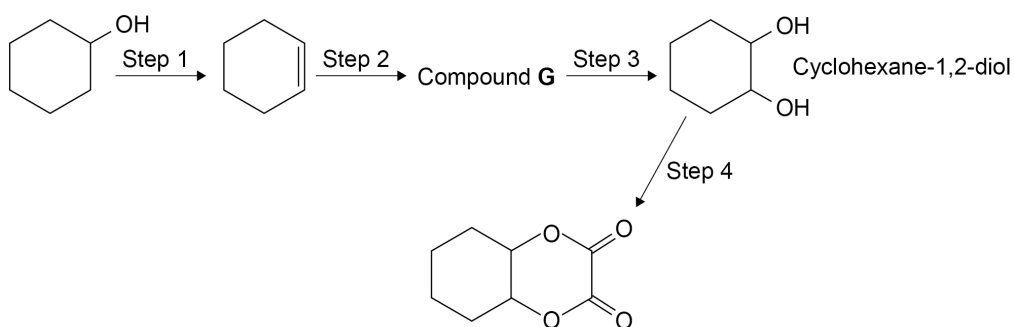


Question	Answers	Additional comments/Guidelines	Mark
06.1	M1 Acylation M2 CH_3COCl OR Ethanoyl chloride M3 AlCl_3 OR Aluminium chloride (mark could be awarded in space for M2)	Allow electrophilic substitution Allow ethanoic anhydride for M2 M3 dependent on M2 Allow Dry/anhydrous for M3 Apply list principle to extra incorrect conditions	1 1 1
06.2	M1 Nucleophilic addition M2 NaBH_4 M3 1-phenyl ethan(-1-)-ol	Allow LiAlH_4 for M2 If H_2/Ni stated allow M2 and M3 but to score a matching M1 it would have to be Catalytic addition	1 1 1

06.3	M1 curly arrow from lone pair to H in H_2SO_4 and arrow from H-O bond to O  M2 protonated intermediate M3 two curly arrows to show loss of water and of H^+ M1 curly arrow from lone pair to H^+  M2 protonated intermediate M3 two curly arrows to show loss of water and of H^+	Penalise M1 for mistakes on structure of H_2SO_4 Allow H^+ attacked in M1 Allow M3 as two steps Allow displayed formulae	3
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0 8

This question is about making a diester from cyclohexanol.

**0 8****1**

State the type of reaction in step 1.

Give the name of the reagent needed for step 1.

[2 marks]

Type of reaction _____

Reagent _____

0 8**2**

State the reagents needed and give equations for step 2 and step 3.

Show the structure of Compound G in your equations.

[4 marks]

Step 2 reagent _____

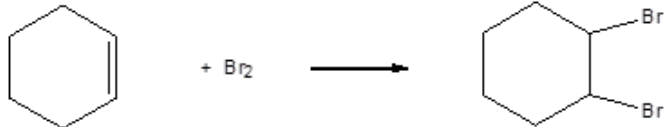
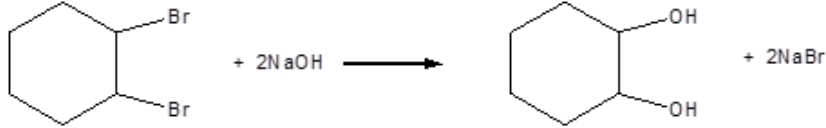
Step 2 equation

Step 3 reagent _____

Step 3 equation



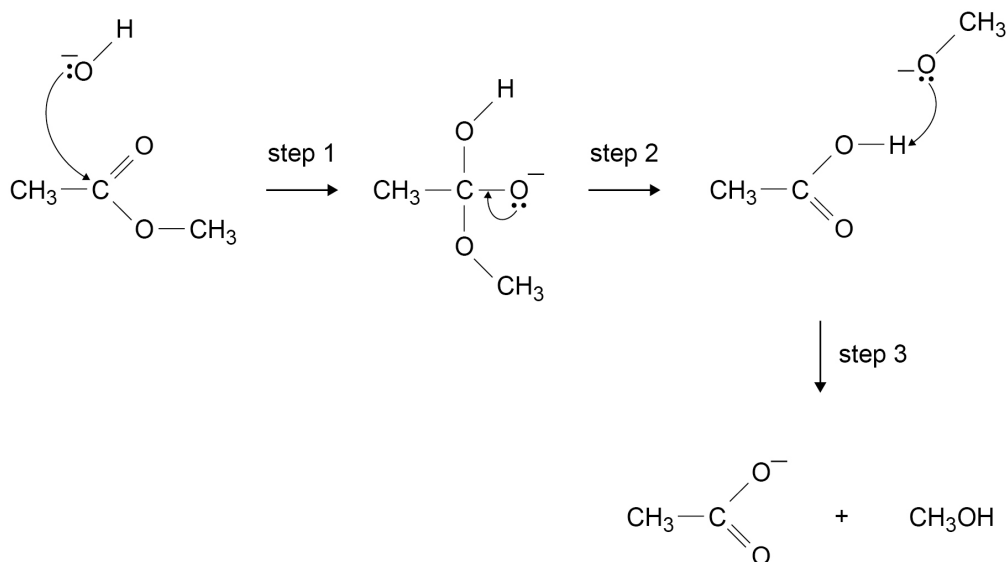
Question	Answers	Additional Comments/Guidelines	Mark
08.1	Dehydration	Allow (acid catalysed) Elimination	M1
	Conc H_2SO_4	Allow Conc H_3PO_4	M2

Question	Answers	Additional Comments/Guidelines	Mark
08.2	Br_2 	Allow bromine (water) Allow Cl_2 or I_2 Allow O_2 if epoxide route used	M1
	NaOH 	allow conseq equation to H_2 , H_2O , HBr , HCl , HI and H_2SO_4 An epoxide is a feasible alternative that could score here and consequentially M3 and M4 Or KOH or other suitable strong alkali	M2
			M3
		Allow this equation with molecular formulae	M4

07

This question is about esters.

Figure 4 shows an incomplete mechanism for the reaction of an ester with aqueous sodium hydroxide.

Figure 4

07.1

Add **three** curly arrows to complete the mechanism in **Figure 4**.**[3 marks]**

07.2

Name the type of reaction shown in **Figure 4**.**[1 mark]**

07.3

Deduce the role of the CH_3O^- ion in step 3 shown in **Figure 4**.**[1 mark]**

07.4

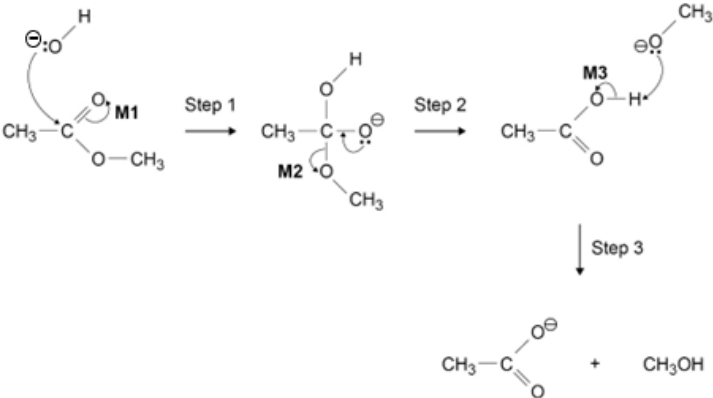
A triester in vegetable oil reacts with sodium hydroxide in a similar way.

Give a use for a product of this reaction.

[1 mark]

Turn over ►



Question	Answers	Additional Comments/Guidelines	Mark
07.1		M1: Arrow from C=O bond to O M2 Arrow from correct C-O bond to O M3 Arrow from O-H bond to O	3 (3 x AO3)

MARK SCHEME – A-LEVEL CHEMISTRY – 7405/2 – JUNE 2022

Question	Answers	Additional Comments/Guidelines	Mark
07.2	(Alkaline/base) hydrolysis		1 (AO1)

Question	Answers	Additional Comments/Guidelines	Mark
07.3	Base	Allow proton acceptor Ignore ref to Bronsted Lowry	1 (AO1)

Question	Answers	Additional Comments/Guidelines	Mark
07.4	Soap only		1 (AO1)

0 8Benzene reacts with methanoyl chloride (HCOCl) in the presence of a catalyst.**0 8 . 1**

Give an equation for the overall reaction when benzene reacts with methanoyl chloride.

Name the organic product.

[2 marks]

Equation _____

Name _____

0 8 . 2

Identify the catalyst needed in this reaction.

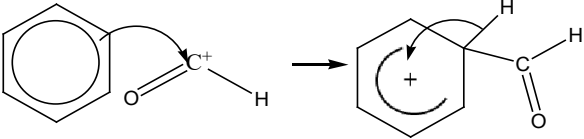
Give an equation to show how the catalyst is used to form the electrophile, $[\text{HCO}]^+$ **[2 marks]**

Catalyst _____

Equation _____

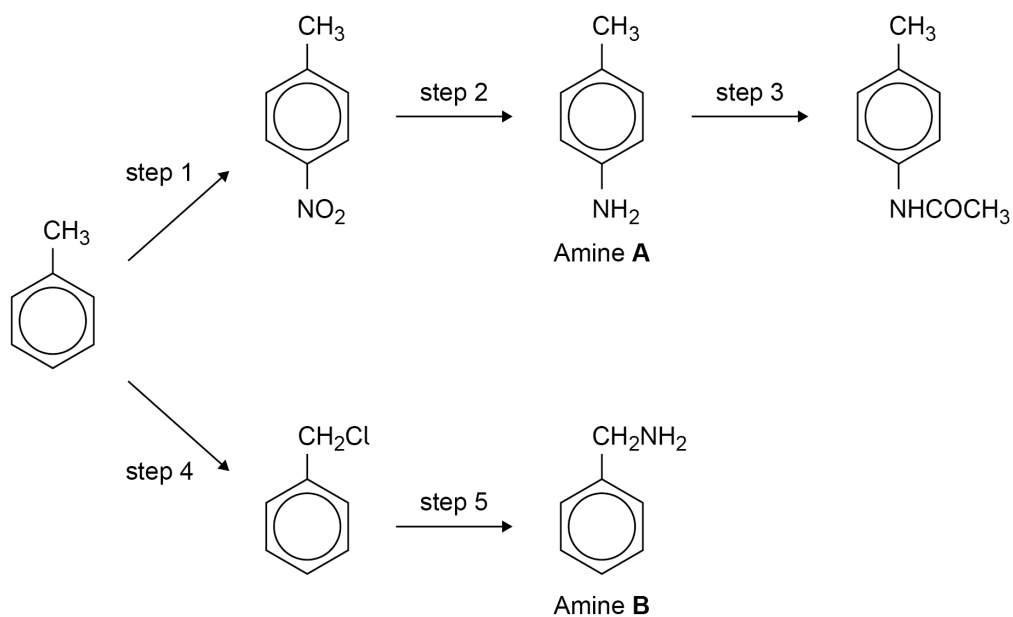
0 8 . 3Outline the mechanism for the reaction of benzene with the electrophile, $[\text{HCO}]^+$ **[3 marks]**

Question	Answers	Additional Comments/Guidelines	Mark
08.1	$\text{C}_6\text{H}_6 + \text{HCOCl} \rightarrow \text{C}_6\text{H}_5\text{CHO} + \text{HCl}$ Or shown as structural formulae Benzaldehyde	Allow phenyl methanal Allow Benzenealdehyde or Benzene carbaldehyde If ethanoyl chloride used allow ecf for name : phenyl ethanone	1 1 (2 x AO2)
Question	Answers	Additional Comments/Guidelines	Mark
08.2	AlCl_3 $\text{HCOCl} + \text{AlCl}_3 \rightarrow [\text{HCO}]^+ + [\text{AlCl}_4]^-$	Allow Aluminium chloride Allow Iron (III) chloride / bromide or formulae Allow + on C or O in equation Can score M1 in equation	1 1 (1 x AO1, 1 x AO2)

Question	Answers	Additional Comments/Guidelines	Mark
08.3		M1 Arrow from inside hexagon to C or + on C M2 Structure of intermediate • horseshoe centred on C1 and must not extend beyond C2 and C6, but can be smaller • + in intermediate not too close to C1 (allow on or “below” a line from C2 to C6) M3 Arrow from bond into hexagon (Unless Kekule) • Can allow M3 arrow independent of wrong M2 structure • + on H in intermediate loses M2 not M3 • Ignore Cl⁻ and AlCl₄⁻ used in M3	M1 M2 M3 (3 x AO2)

1 0

This question is about the reaction scheme shown.



1 0 . 1

State the reagents needed for step 1 and the reagents needed for step 2.

[3 marks]

step 1 _____

step 2 _____

1 0 . 2

Give the name of the mechanism for the reaction in step 3.

[1 mark]



1 0 . 3 Name the reagent for step 4.

State a necessary condition for step 4.

[2 marks]

Reagent _____

Condition _____

1 0 . 4 Amine **A** is formed in step 2 and amine **B** is formed in step 5.

Explain why the yield of **B** in step 5 is less than the yield of **A** in step 2.

[2 marks]

1 0 . 5 Explain why amine **B** is a stronger base than amine **A**.

[2 marks]

10

END OF QUESTIONS



Question	Answers	Additional Comments/Guidelines	Mark
10.1	Step 1 Conc HNO ₃		M1
	Step 1 Conc H ₂ SO ₄	If conc missing in both allow 1 for HNO ₃ and H ₂ SO ₄	M2
	Step 2 Sn and HCl	Allow Fe and HCl or Ni and H ₂	M3 (3 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
10.2	(nucleophilic) addition-elimination		1 (AO1)

Question	Answers	Additional Comments/Guidelines	Mark
10.3	Chlorine	Allow Cl ₂	M1
	UV (light)	Allow sunlight / High temp (above 300°C)	M2 (2 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
10.4	In Step 5 further substitution / gives other amine products		1
	In Step 2 only one amine		1 (2 x AO3)

Question	Answers	Additional Comments/Guidelines	Mark
10.5	In B Alkyl group is electron donating or positive inductive effect Lone pair <u>on N</u> more available	Or in A lone pair (on N partially) delocalised Lone pair <u>on N</u> less available	1 1 (2 x AO2)